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Effective and Explainable Molecular Property Prediction by Chain-of-Thought Enabled Large Language Models and Multi-Modal Molecular Information Fusion

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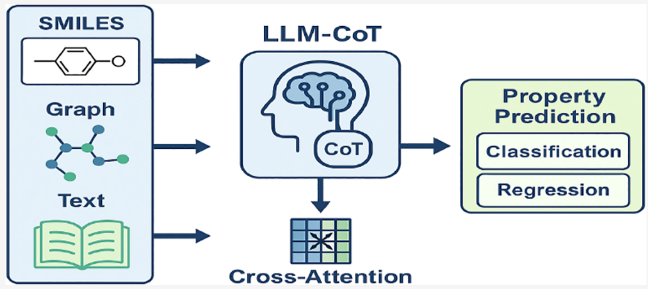
ABSTRACT: Molecular property prediction (MPP) plays a critical role in drug design and discovery. Due to the multimodal nature of molecular data (e.g., 1D SMILES strings and 2D molecular graphs), multimodal information fusion can generally achieve better performance than using single-modality molecular data. On the other hand, with the rise of large language models (LLMs), increasing efforts have been made to leverage LLMs for molecular property prediction. However, existing works usually use one or two modalities of molecular data, employ simple techniques for multimodal information integration such as straightforward concatenation and summation, and cannot comprehensively exploit the rich complementary information across multiple modalities. Furthermore, these models are typically designed for general chemical tasks, making their performance in molecular property prediction suboptimal. Worse still, they cannot provide explainable results, which are important for drug design and discovery-related tasks including MPP. In response to these limitations of existing works, this paper presents LLM-MPP, a new, effective, and explainable LLM-driven multimodal method for drug molecular property prediction, which leverages 1D SMILES strings, 2D molecular graph structures, and molecular textual descriptions of molecular properties as training data. By incorporating the chain-of-thought (CoT) technique, we enhance the interpretability and transparency of the proposed method, while promoting alignment and feature extraction across multiple modalities. Cross-attention and contrastive learning are adopted to effectively fuse multimodal molecular representations for property prediction. Experiments on nine benchmark data sets for molecular property prediction demonstrate that our method achieves state-of-the-art performance on 5 and ranks second on 1 of the 9 data sets, surpassing 22 existing baselines. Ablation experiments validate the effectiveness of our innovative modules, effectively addressing the limitations of existing models.



■ INTRODUCTION

Molecular property prediction (MPP) is a fundamental task in drug design and discovery,[1] aiming to predict various characteristics of newly designed/discovered drug candidates, including biological activity, physicochemical properties, and ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) properties.[2,3] Traditionally, MPP relies on molecular and cellular assays and molecular docking simulations. However, compared to these conventional drug development practices, methods based on machine learning (ML) and particularly deep learning (DL), offer substantial benefits[4−6] such as shorter development timelines, lower financial costs, and higher success rates. As a result, in recent years, there has been a growing body of research that employs ML/DL techniques to boost the prediction of drug-molecule properties.

Molecular data are inherently multimodal, consisting of various types of representation forms that offer complementary insights into molecular structures and characteristics. These representing forms include 1D strings, 2D molecular graphs, and 3D spatial structures, each of which requires specialized deep learning techniques for effective modeling. Typical 1D forms represent molecules as strings such as SMILES (The Simplified Molecular Input Line Entry System)[7] and IUPAC,[8] encoding molecules in a linear, text-like format. Natural language processing (NLP) models[9−11] are well suited to process these strings, as they excel at extracting semantics and learning the syntax and grammar rules implied in 1D text-like strings, allowing them to understand the underlying chemical properties of molecules. 2D forms are mainly molecular graphs, where atoms are represented as nodes and chemical bonds as edges. Graph neural networks[12−16] are the standard approaches to

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learning the graphic structures of molecules, as they can effectively model connectivity and chemical relationships within molecules. Typical graph models include message passing neural networks,[12] graph convolutional networks[14] and graph attention networks.[15] 3D spatial structures of molecules provide crucial information about the geometric arrangement of atoms, and geometric deep learning models such as SchNet[17] and DimeNet[18] have been proposed to learn from 3D molecular coordinates. There are also other modalities of molecular data such as molecular fingerprints,[19] molecular descriptors,[20] and textual descriptions, which play important roles in understanding molecules.

A single modality is often insufficient to capture the complex nature of molecules. In contrast, multiple modalities can provide more comprehensive and complementary information, enhancing the overall quality of molecular representations. Meanwhile, large language models (LLMs) such as GPT,[21,22] LLaMA[23] and recently DeepSeek[24] have emerged as powerful tools in various domains. Their great success in tasks such as natural language understanding[25] and generation[23] have inspired researchers to explore their potential in molecular property prediction.[26−30]

Despite these advances, existing LLM-based molecular property prediction approaches have obvious limitations. First, current LLM-based models typically use only 1D molecular data,[31] or integrate only two modalities of molecular data, such as combining 1D SMILES strings and 2D graphs[28] or 3D coordinates,[29] without fully exploiting richer multimodal data. And their fusion techniques are generally simple, limiting their ability to generate robust and well-aligned multimodal representations. Second, these models lack interpretability,[32] making them difficult to trace the reasoning process behind any prediction, which is a severe drawback for high-stake applications like drug discovery. Third, most LLM-based models are designed for general chemical tasks,[28−30,33] targeting a wide range of applications such as molecule description generation, chemical entity recognition, and other fundamental chemical tasks, rather than being optimized specifically for property prediction, which results in suboptimal performance of molecular property prediction.

To address these limitations, in this paper we propose a novel method LLM-MPP that leverages LLMs for molecular property prediction. On the one hand, we propose a multimodal fusion strategy that integrates three distinct modalities of molecular information including 1D SMILES strings, 2D molecular graphs, and textual descriptions of molecules. We utilize LLMs to extract and align features from 1D SMILES strings and textual descriptions, providing richer supplementary information for molecular representations. Additionally, we employ graph neural networks to represent molecular graphs. Then, we adopt cross-attention and contrastive learning to align and fuse representations from different modalities, ensuring that structural information and semantic information are effectively combined. This leads to more robust and well-aligned multimodal representations that boost model performance. On the other hand, to enhance the interpretability of the model, we introduce chain-of-thought (CoT) prompting[34] to guide the LLM to perform step-by-step reasoning when predicting the properties. By simulating the reasoning process, CoT enhances the transparency of the model, allowing users to trace the underlying logic behind each prediction.

In summary, our contributions are as follows.

1. We propose a new effective and explainable MPP method LLM-MPP, which adopts large language models and fuses multimodal molecular data, including 1D SMILES strings, 2D molecular graphs, and textual descriptions.
2. We employ cross attention and contrastive learning to fuse the multimodal molecular data, which provides richer supplementary information of molecules for property prediction.
3. We empower interpretability in the model by introducing a chain-of-thought technique, which guides the LLM to generate explicit reasoning steps, thus making the model's predictions more transparent.
4. We conduct extensive experiments on 9 diverse benchmark data sets from MoleculeNet. Experimental results show that our LLM-based model LLM-MPP achieves outstanding performance, superior to 22 existing models on five data sets and ranks second on one additional data set.

■ RELATED WORK

Multi-Modal Representation Learning for Molecular Property Prediction. With the advancement of deep learning, integrating various modalities of molecular data has become the main method for improving the accuracy of molecular property prediction.

The fusion of 1D molecular strings and 2D molecular graphs is a common approach. For example, GraSeq[35] combines 1D SMILES molecular sequences with 2D molecular graphs, using Bi-LSTM networks[9] to encode SMILES sequences and graph convolutional networks[14] to encode molecular graphs, and then fuse them by weighting sum to obtain unified molecular representations. Xie et al.[36] proposed a chemistry-aware molecular fragment prediction method within a self-supervised contrastive learning framework, which innovatively utilizes fragment-based molecular graphs to represent the topological relationships among chemically aware substructures, and employs contrastive learning between SMILES and graph representations at the fragment level for pretraining. For fusing 2D molecular graphs and 3D spatial structures, Luo et al.[37] claimed that transformer models[10] can simultaneously understand both 2D and 3D molecular representations. They developed Transformer-M, which takes 2D and 3D molecular data as input to generate meaningful semantic representations. GraphMVP[38] employs self-supervised learning using the alignment and consistency between the 2D topological structures and their corresponding 3D geometric representations. Jiang et al.[39] proposed the MultiGran-SMILES model, which represents molecules at three levels of granularity, i.e., atom level, molecular substructure and molecular graph, and employs BiGRU[40] to encode SMILES sequences at the atomic and substructure levels, while GraphSage[16] is used to encode molecular graphs. Besides, Chen et al.[41] proposed the AGBT framework, which integrates 1D SMILES-based transformer representations with 3D structural information encoded via algebraic graphs.

Large Language Models in Chemistry and Drug Discovery. In recent years, large language models (LLMs) have gained increasing attention in the fields of chemistry and drug discovery. These models are trained to perform tasks such as molecule generation, chemical entity recognition, reaction prediction, and molecular property prediction. Guo et al.[30] established a comprehensive benchmark containing eight

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Table 1. Comparison of Models for Molecular Property Prediction[a]

model	data modality	fusion method	interpretability
		single-modal models	
ST[44]	1D	–	low
AttentiveFP[45]	2D	–	low
SphereNet[46]	3D	–	low
		simple MM models	
GraSeq[35]	1D+2D	a simple fusion layer	low
GraphMVP[38]	2D+3D	self-supervised learning	low
		LLM-based models	
LLaMA-2–7B chat[23]	1D	–	low (black-box approach)
InstructMol[28]	1D+2D	concatenation in the prompt	low
Graph2Token[43]	1D+2D	concatenation and alignment in the prompt	low
LLM-MPP (ours)	1D+2D+texts	prompting+CL+CA	high (CoT reasoning)

[a]Here, MM: Multi-Modal, MPP: Molecular property prediction, 1D: 1D SMILES strings, 2D: 2D molecular graphs, 3D: 3D molecular spatial structures, ST: SMILES transformer model, CL: contrastive learning, CA: cross attention, "–": unavailable.

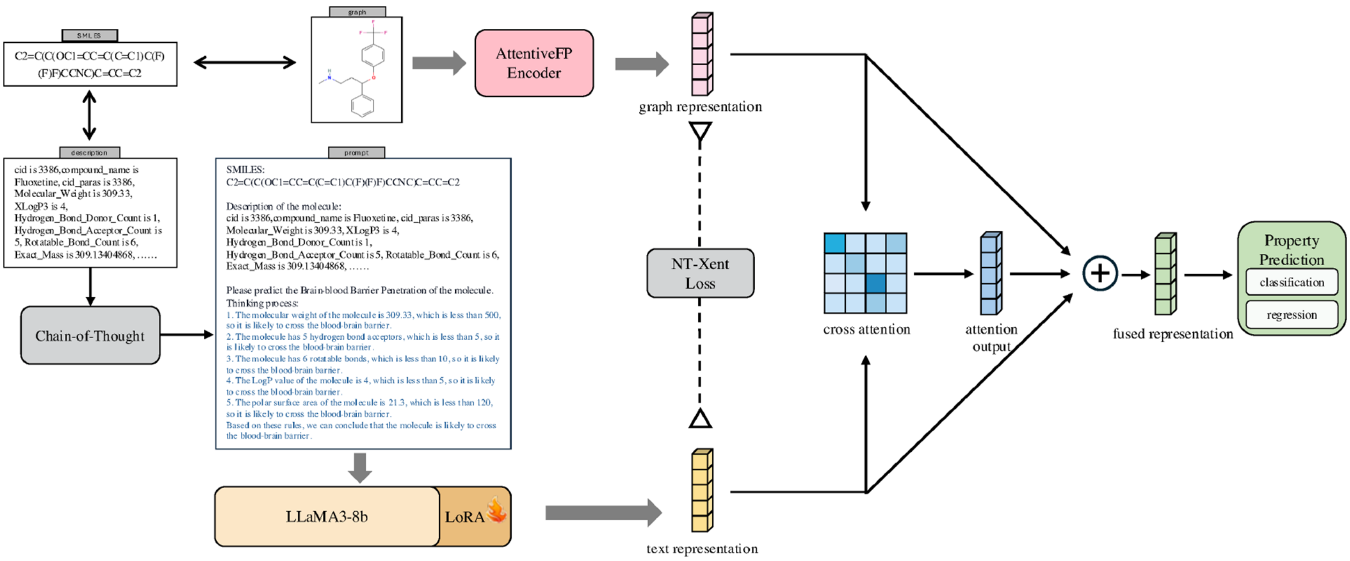

Figure 1. Framework of LLM-MPP. ⊕ in the figure represents the weighted representation aggregation operation.

chemistry tasks, facilitating a wide exploration of the capabilities of LLMs in the field of practical chemistry. Liang et al.[42] developed a prototype system DrugChat that shows ChatGPT-like capability on drug molecule graphs. Fang et al.[33] presented a comprehensive instruction data set Mol-instructions, covering molecule-oriented instructions, protein-oriented instructions, and biomolecular text instructions. InstructMol[28] is a multi-modal LLM aligning molecular structures with natural language using a two-stage training strategy. Graph2Token[43] is an efficient model that aligns graph tokens to LLM tokens. Li et al.[29] leveraged large language models to integrate multiple molecular modalities, including 1D SMILES strings, textual descriptions, and 3D coordinate information. Zhang et al.[32] proposed a model called MolGraph-LarDo that integrates SMILES, molecular graphs, and LLM-generated descriptions by a two-stage prompting method for molecular property prediction.

Differences between Our Work and Existing Methods. Current approaches to fusing and aligning multimodal molecular representations often use overly simple techniques such as concatenation or summation of features,[35,39] which limit their abilities to fully exploit the complementary information provided by diverse molecular modalities. There is a pressing need for more advanced fusion techniques that can effectively integrate more molecular modalities and align their features. For current large language models in molecular property prediction, they usually focus on single-modality data or integrate only two modalities, such as 1D SMILES strings with 2D[28] molecular graphs. Most LLMs in the chemistry field are developed for general tasks,[28–30,33] rather than specifically tailored for drug molecular property prediction. Consequently, their performance tends to be suboptimal and often fails to capture the nuanced chemical characteristics essential for accurate molecular property prediction. Additionally, a severe drawback of many LLM-based models is their lack of interpretability. The inability to trace the reasoning behind predictions makes these models difficult to apply in high-stake applications like drug discovery. For example, although MolGraph-LarDo proposed by Zhang et al.[32] uses a two-stage prompting method, its prompting strategy focuses on description generation and lacks reasoning mechanisms.

Differently, our work addresses these issues by introducing a multimodal fusion framework based on large language models optimized specifically for molecular property prediction. We integrate three types of molecular data including 1D SMILES strings, 2D molecular graphs, and textual descriptions, and

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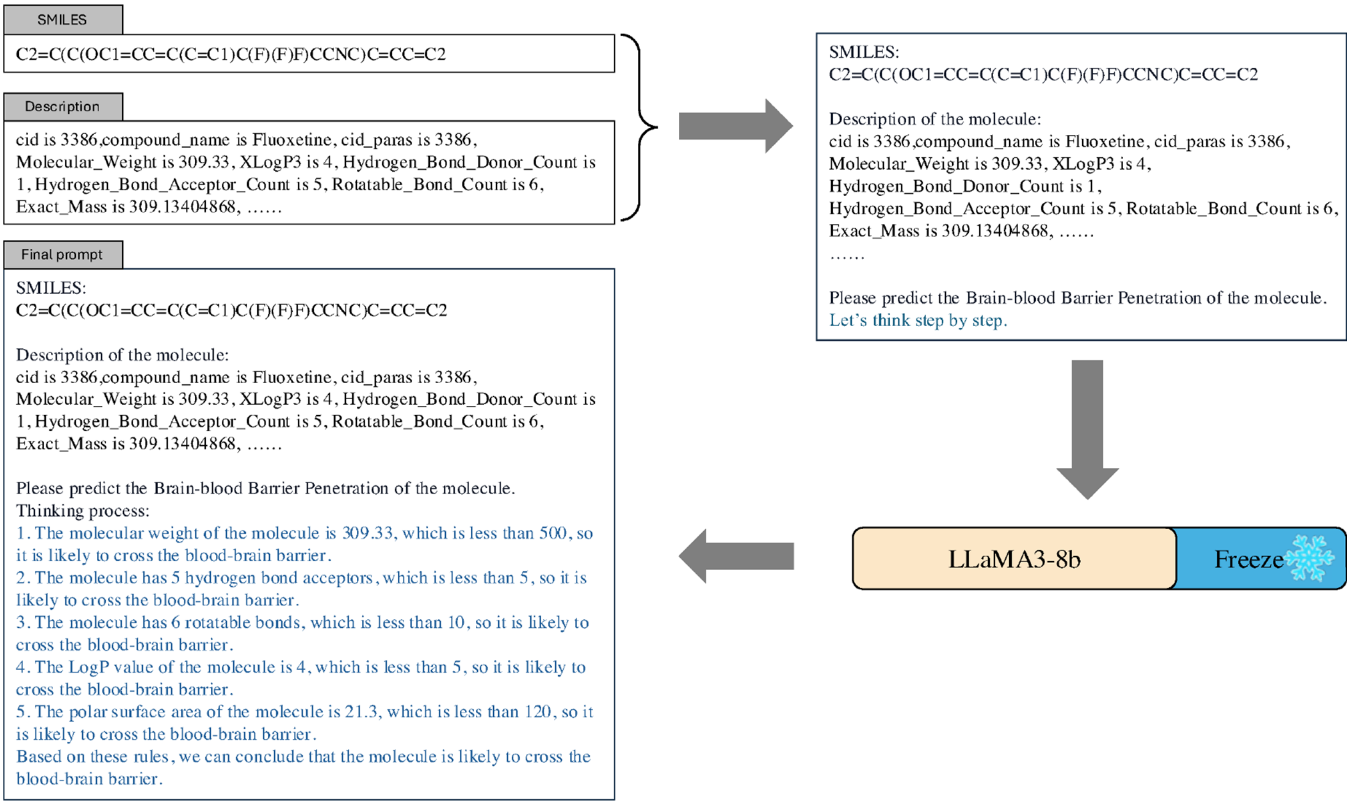

Figure 2. Process of chain-of-thought prompting.

employ advanced fusion techniques such as cross-attention to generate high-quality, well-aligned representations. Additionally, we enhance the model's interpretability through chain-of-thought prompting, which makes the reasoning behind predictions more transparent and easier to understand. Finally, our model is specifically fine-tuned for property prediction, overcoming the limitations of current LLM-based methods in drug molecular property prediction.

To concisely highlight the differences between our method and existing works, Table 1 compares our model with different types (single-modal models, simple multimodal models, and LLM-based models) of current models for molecular property prediction from three perspectives, i.e., *data modality*: how many modalities of molecular data are used? *fusion method*: what technique is used for multimodal data fusion? and *interpretability*: whether the model is interpretable?

■ METHODS

Framework. Figure 1 illustrates the framework of our method LLM-MPP. LLM-MPP utilizes three sources of data, including 1D SMILES strings, 2D molecular graphs, and textual descriptions of molecular properties. Concretely, we design a novel prompt that incorporates the SMILES strings, textual descriptions and texts generated by chain-of-thought (CoT) prompting, which is fed into a large language model (LLM) for text embedding and representation learning. This approach enhances the LLM's understanding of molecular structures and properties, facilitating molecular semantic understanding, and reducing hallucination. For molecular graphs, a graph neural network, AttentiveFP,[45] is used to encode the molecular graph structures, producing graph-based representations. A cross-attention mechanism is then employed to fuse the text and graph representations, which are subsequently combined via weighted sum to obtain the final fused representation. The fused embeddings are used for downstream molecular property prediction. Additionally, contrastive learning is applied to reinforce the consistency between the text and graph representations of molecules, leading to more robust molecular representations. Further technical details of LLM-MPP are elaborated in the following sections.

LLM for Molecular Text Representation Learning with Chain-of-Thought. *Chain-of-thought for Molecular Understanding.* To enhance the LLM's ability to understand molecular SMILES strings and textual descriptions, we introduce a chain-of-thought (CoT) prompting strategy. This technique encourages the LLM to simulate a step-by-step reasoning process, enabling it to decompose and analyze various molecular characteristics in a structured manner. Figure 2 illustrates the process of the chain-of-thought technique. We initially design a prompt containing the SMILES strings and textual descriptions of the molecules, and append the instruction "Let's think step by step" at the end to guide the reasoning process. This prompt is then fed into a frozen LLM, which generates text of the step-by-step thought process. Concretely, LLM-MPP leverages LLaMA3−8b, a powerful LLM that is pretrained on vast amounts of general and scientific textual data. This allows the LLM to interpret and extract meaningful insights from molecular SMILES strings and textual descriptions.

The generated step-by-step text can not only guide the subsequent prediction reasoning process, but also enhance the model's interpretability, as the reasoning path can be traced to understand what factors contribute to the final prediction.

Novel Prompt Integrating Molecular SMILES Strings, Property Textual Descriptions and CoT Reasoning Texts. To effectively align and utilize molecular SMILES strings, molecular textual descriptions, and the text of reasoning process generated

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by chain-of-thought, we design a unique prompting method that integrates all this information as input to an LLM. Figure 3 illustrates the complete template of our proposed prompt, and an example of the prompt is shown in Figure 1. The prompt combines the SMILES string of a molecule with relevant textual description, detailing properties such as molecular weight, bond counts, and heavy-atom counts, followed by the reasoning text generated by chain-of-thought prompting, which is presented in a structured format. The SMILES string and textual description guide the LLM to understand the relationship between molecular structure and properties, and jointly process structural and property-related information. This enriched prompt improves the model's ability to generate precise text representations and align the information extracted from SMILES and text descriptions, and improves the interpretability and transparency of the model.

SMILES:
<SMILES string of the molecule>

Description of the molecule:
<Description of the molecule>

Please predict the <property name> of the molecule.
Thinking process:
<Text generated by chain-of-thought>

Figure 3. Prompt template for LoRA fine-tuning. Blue texts in the first and second angle brackets represent content filled in according to specific molecules from the data set. Blue text in the third angle brackets indicates the text of reasoning process generated by CoT.

To further adapt the LLM to the molecular property prediction task, we apply Low-Rank Adaptation (LoRA) fine-tuning[47] as shown in Figure 1. LoRA is a parameter-efficient fine-tuning method that allows the model to be modified only a small subset of the parameters, thus significantly reducing computational cost and the risk of overfitting. By fine-tuning the LLM with LoRA, we enhance its ability to capture molecular information, ensuring that the LLM retains its general prior knowledge while focuses on extracting chemical and molecular features. We extract the hidden embedding of the final layer from the LLM's output as the molecular text representation.

Graph Neural Network for Molecular Graph Representation. To represent the molecular graphs, we employ a Graph Neural Network (GNN) model, specifically AttentiveFP.[45] Figure 4 illustrates the structure of the AttentiveFP model, which comprises the input layer, atom embedding layers, molecule embedding layers, and task-specific layers. The input layer encodes the initial features of atoms and edges for further processing. The atom embedding layers utilize an attention mechanism to aggregate features from neighboring nodes via message passing. Specifically, the aggregation mechanism uses attention weights to determine the importance of each neighboring node, followed by a gated recurrent unit (GRU) that updates the state of each atom, transitioning from the previous state to the current one through a readout process. In the molecule embedding layers, the entire molecule is represented as a super virtual node that connects all the atoms within the molecule. This super node is embedded using the same attention-based atom embedding mechanism. The model

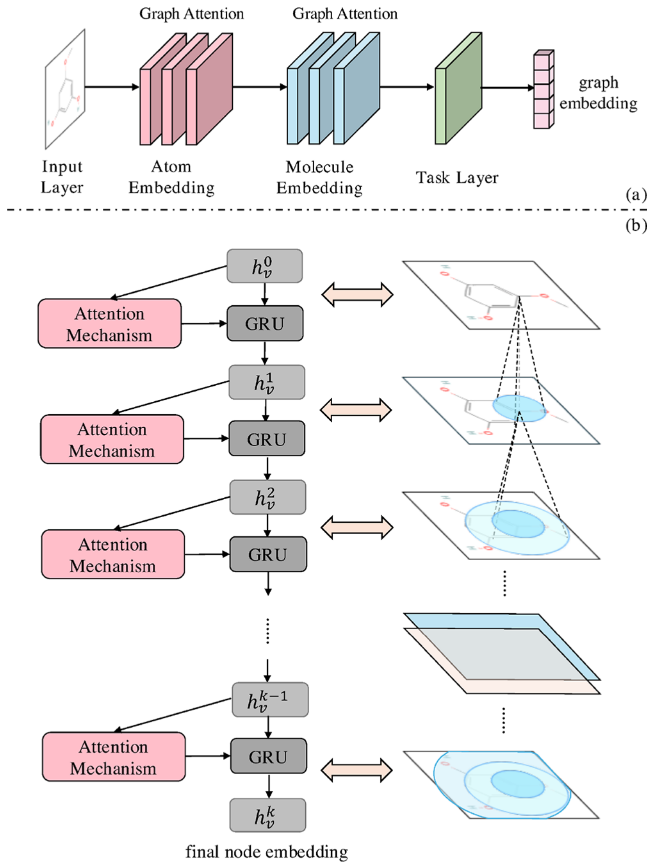

Figure 4. (a) The architecture of the AttentiveFP[45] module. (b) The detailed process of an atom embedding module.

iteratively refines the molecular embeddings through multiple layers of graph attention, ultimately generating a comprehensive state vector for the entire molecule. These final graph embeddings serve as the input for downstream tasks, where they are fused with text-based representations for effective prediction of molecular properties.

The message passing is formulated as

$$C_v^{k-1} = \sum_{u \in N(v)} M^{k-1}(h_u^{k-1}, h_v^{k-1}) \tag{1}$$

where $N(v)$ represents all of the neighbors of node v, and M^{k-1}, h_u^{k-1} and h_v^{k-1} refer to the message function (specifically graph attention), the hidden representations of node u and node v, respectively at iteration $k - 1$.

Formally, the readout phase is

$$h_v^k = \mathrm{GRU}^{k-1}(C_v^{k-1}, h_v^{k-1}) \tag{2}$$

where GRU^{k-1} represents the update function GRU, which is a gated recurrent unit at iteration $k - 1$.

Text-Graph Representation Fusion. To effectively leverage the complementary information of textual representations and graph representations, we propose a fusion method consisting of two steps: cross-attention fusion and weighted representation aggregation.

Cross-Attention Fusion. The cross-attention fusion module is designed to align and integrate text and graph representations dynamically, capturing the interdependencies between these two modalities with a multihead cross-attention mechanism, which effectively adjusts the molecular text representations in

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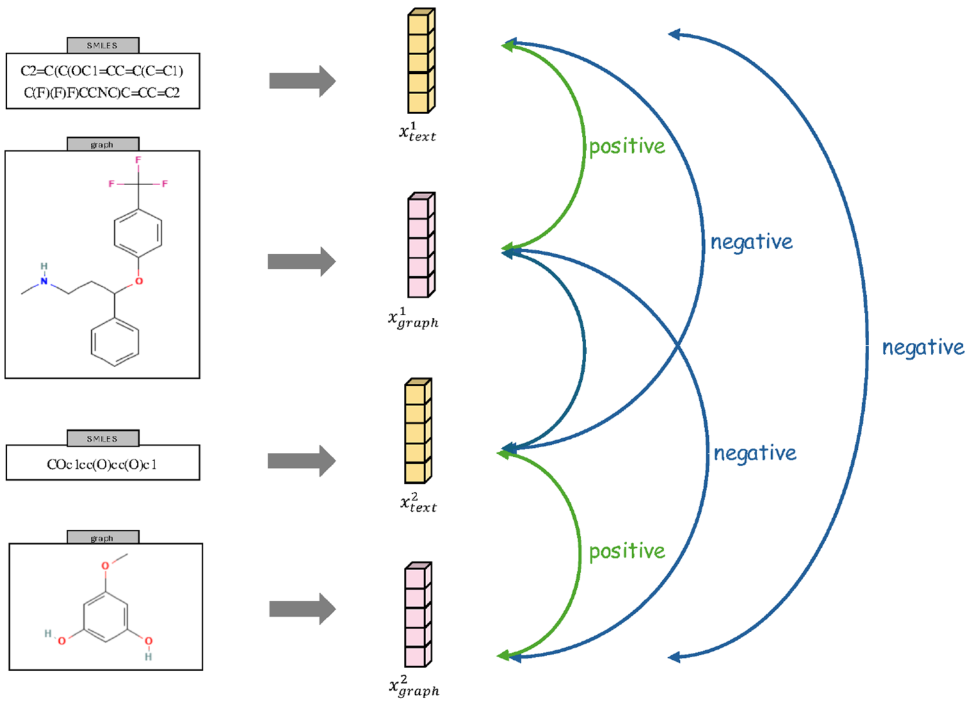

Figure 5. Illustration of contrastive learning. x^1_{text}, x^1_{graph}, x^2_{text}, x^2_{graph} represent the text representations and graph representations of two different molecules, respectively.

response to the information provided by the graph representations. The module begins by projecting the key and value components of the text representations to match the dimensionality of the graph-based queries. Specifically, we use a linear transformation to adjust the dimensions, ensuring compatibility with attention computation. We then apply a multihead attention mechanism, which divides the representations into multiple attention heads. Formally,

$$\begin{aligned} K &= W_k X_{\text{text}} \\ V &= W_v X_{\text{text}} \\ Q &= W_q X_{\text{graph}} \\ h_n &= \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \\ X_{att} &= \text{MultiheadAttention}(Q, K, V) \\ &= \text{Concat}(h_1, h_2, \cdots, h_m)W_O \end{aligned} \tag{3}$$

where X_{text}, X_{graph}, X_{att} are text representations, graph representations and attention outputs of a batch of molecules. W_k, W_v, W_q are learnable matrices that project text and graph representations into the query, key, and value spaces of the same dimension. Each h_n is the independent attention output corresponding to an attention head, and m is the number of attention heads. W_O is a learnable output projection matrix.

Weighted Representation Aggregation. We employ a weighted summation strategy to combine the cross-attention output with the original embeddings. This method allows for the integration of features, giving more importance to the modality that provides more relevant information for the downstream task. Formally, the weighted sum of representations can be represented as

$$x_{\text{fused}} = \eta_t x_{\text{text}} + \eta_g x_{\text{graph}} + (1 - \eta_t - \eta_g) x_{\text{att}} \tag{4}$$

where x_{fused}, x_{text}, x_{graph} denote the final fused representation, the text representation, and the graph representation of the molecule, respectively. η_t and η_g are hyperparameters that indicate the importance of the text and graph representations in the aggregation module.

Contrastive Learning Loss. Figure 5 shows the process of contrastive learning. The objective of contrastive learning is to learn discriminative representations by narrowing the gap between "similar" positive pairs and enlarging the distance between "dissimilar" negative pairs in the latent space. Specifically, we employ the Normalized Temperature-Scaled Cross-Entropy Loss (NT-Xent Loss) in LLM-MPP. We define the positive pairs as the representations of the same molecule derived from two different modalities: one from the text representations and the other from the graph representations. These pairs are treated as "similar" and are encouraged to have a high degree of similarity in the latent space. On the other hand, the negative pairs are composed of representations from different molecules, which are considered "dissimilar" and are encouraged to be as distinct as possible in the latent space. We use cosine similarity to measure the similarity between two vectors of representations,

$$\text{sim}(z, z') = \frac{z^T z'}{\|z\|_2 \|z'\|_2} \tag{5}$$

where z and z' are two vectors used in the contrastive learning.

Therefore, the NT-Xent loss in LLM-MPP is as follows:where x_i and x_j represent the text and graph representations of the same

$$\begin{aligned} \mathcal{L}_{\text{NT-Xent},i,j} &= -\log \frac{S(x_i, x_j)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} S(x_i, x_k)} \\ S(x_i, x_j) &= e^{sim(x_i, x_j)/\tau} \\ \mathcal{L}_{\text{NT-Xent}} &= \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_{\text{NT-Xent},i,j} \end{aligned} \tag{6}$$

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molecule, respectively, and N is the number of molecules in the batch. x_k refers to any representation of a molecule in the batch, including both text and graph representations, except for x_i. τ is a hyperparameter to control the sharpness of the similarity distribution. In the NT-Xent loss, (x_i, x_j) constitutes a positive pair, while (x_i, x_k) denotes a negative pair.

This contrastive learning approach improves the model's ability to learn robust and discriminative features from both text and graph data, ensuring that multimodal representations are well aligned and effective for downstream tasks such as prediction of molecular properties.

The total loss of LLM-MPP is a weighted combination of the NT-Xent contrastive loss and the property prediction loss, i.e.,

$$\mathcal{L} = \mathcal{L}_{\text{prediction}} + \lambda \mathcal{L}_{\text{NT-Xent}} \tag{7}$$

where λ is a weighting factor that balances the contribution of the contrastive loss and the property prediction loss.

Prediction. We use an MLP network for the final prediction, which consists of three linear layers, and the first two layers are each followed by a ReLU activation function. Formally,

$$\begin{aligned} o_1 &= \text{ReLU}(\text{Linear}(x_{\text{fused}})) \\ o_2 &= \text{ReLU}(\text{Linear}(o_1)) \\ O &= \text{Linear}(o_2) \end{aligned} \tag{8}$$

where O represents the predicted result of the classification or regression task.

EXPERIMENTS AND RESULTS

In this section, we evaluate LLM-MPP against a number of state-of-the-art methods on 9 public benchmark data sets. We first provide a brief introduction to the data sets used and the baselines compared. Then, we describe the experimental setup. Finally, we present experimental results, including the performance comparison of our model with the compared methods on the benchmark data sets, results of ablation studies, visualization of molecular representations, examples of texts generated by chain-of-thought, and a case study of several real drugs predicted using our model.

Data Sets. *Benchmark Data Sets.* We conduct experiments on 9 benchmark data sets from MoleculeNet,[48] including five classification data sets and four regression data sets. Table 2 presents the statistics of the 9 data sets, including the numbers of tasks and molecules, task type and performance metric. Details of the data sets are as follows:

- The BBBP data set classifies compounds as permeable or nonpermeable to the blood-brain barrier.
- The BACE data set provides qualitative binding data for human β-secretase inhibitors.
- The ClinTox data set consists of FDA-approved drugs along with those that failed in clinical trials due to toxicity issues.
- The SIDER data set contains the side effects of drugs.
- The Tox21 data set offers toxicity data evaluated for 12 different toxicity targets.
- The FreeSolv data set contains experimental and calculated hydration free energy of small molecules in water.
- The ESOL data set provides water solubility data.
- The Lipo data set includes experimental results of octanol/water distribution coefficient of 4200 compounds.
- The QM7 data set records electronic properties obtained through DFT.

For all the data sets, we follow the standard data split protocol recommended by MoleculeNet,[48] using an 8:1:1 ratio for training, validation, and test sets.

Molecular Text Descriptions for Data Enhancement. We enhance the data sets by incorporating additional textual descriptions of molecular properties obtained from the PubChem database.[49] For the molecules in these data sets that have property values in PubChem, we used the compound identifier (CID) to retrieve the properties directly. We ensure that the data collection process is in compliance with the licensing requirements and used the open interface provided by PubChem for data access.

We consider 26 major physicochemical properties of molecules in PubChem. Concretely, they are Molecular Weight, XLogP3, Hydrogen Bond Donor Count, Hydrogen Bond Acceptor Count, Rotatable Bond Count, Exact Mass, Monoisotopic Mass, Topological Polar Surface Area, Heavy Atom Count, Formal Charge, Complexity, Isotope Atom Count, Defined Atom Stereocenter Count, Undefined Atom Stereocenter Count, Defined Bond Stereocenter Count, Undefined Bond Stereocenter Count, Covalently-Bonded Unit Count, Canonicalization Status, Physical Description, Color/Form, Odor, Boiling Point, Melting Point, Flash Point, Solubility, and Density.

We transform these physicochemical property values into structured textual descriptions. Specifically, for each property of a molecule, we describe the property by the format as follows:

[Property name] is [specific value]

Here, [Property name] refers to any of the 26 properties above, and [specific value] means the actual value extracted from PubChem. This process above generates textual descriptions for the available properties of each molecule. For molecules not available in PubChem, the description fields are left empty.

The inclusion of textual property descriptions enables more robust learning of molecular representations, allowing better predictive modeling of molecular behaviors and interactions. These textual descriptions do not cause data leakage in the MPP tasks, as they do not contain any of the target properties of the MPP tasks. Concretely, we masked all text fragments that explicitly mention or imply target properties. For example, for the ESOL data set, we removed the "solubility" property information in textual descriptions from PubChem. This ensures that the large language model cannot access or infer the target labels directly from the input prompt, thereby avoiding information leakage and preserving the integrity of the

Table 2. Statistics of Benchmark Datasets Used in Our Experiments

data set	#tasks	#molecules	task type	metric
BBBP	1	2039	classification	ROC-AUC
BACE	1	1513	classification	ROC-AUC
ClinTox	2	1478	classification	ROC-AUC
SIDER	27	1427	classification	ROC-AUC
Tox21	12	7831	classification	ROC-AUC
FreeSolv	1	642	regression	RMSE
ESOL	1	1128	regression	RMSE
Lipo	1	4200	regression	RMSE
QM7	1	6830	regression	MAE

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evaluation. Besides, from mathematical perspective, representations produced by LLMs are in high-dimensional spaces (e.g., 4096-dimension in LLaMA3), where randomly sampled or uncorrelated inputs tend to be approximately orthogonal due to the concentration of measure. As a result, the high-dimensional embeddings of textual features are unlikely to align with the target label directions, which mitigates the risk of implicit data leakage.

On the other hand, although large language models (LLMs) such as LLaMA3 are pretrained on extensive public corpora, it is possible that some molecule-related information from the test data sets might be used during pretraining. However, general molecular descriptions alone do not constitute data leakage unless the model is exposed to the explicit mapping between a specific molecule and its corresponding property label.

To verify whether the observed performance improvements arise from our proposed multimodal framework rather than from prior knowledge memorized during pretraining, we conducted a controlled ablation experiment. Specifically, we froze the parameters of the LLM and provided it with only the SMILES strings of the molecules as input, without molecular descriptions, chain-of-thought prompts, and graph structures. This experimental setup tests whether the LLM, relying solely on SMILES representations, can accurately predict molecular properties:

- If the model achieves high performance in this setting, it suggests possible memorization of molecule-label pairs from the pretraining corpus.
- Conversely, if the model performs poorly, it indicates that the performance gains in our full model stem from effective multimodal feature extraction and integration.

We performed this experiment on two classification data sets (BACE and BBBP) and two regression data sets (ESOL and Lipo). The results, which are summarized in Table 3, demonstrate that freezing the LLM and removing auxiliary modalities lead to significant performance drop, confirming the necessity and effectiveness of our multimodal learning design.

Table 3. Performance Comparison on Classification (ROC-AUC ↑) and Regression (RMSE ↓) Datasets[a]

model	classification (AUC ↑)		regression (RMSE ↓)	
	BACE	BBBP	ESOL	Lipo
LLM-ablation	0.396	0.645	3.446	2.309
LLM-MPP (ours)	**0.874**	**0.973**	**0.829**	**0.664**

[a]The best result is marked in **bold**. LLM-ablation represents the ablation setting mentioned above.

Baselines. We compare our method LLM-MPP with 22 existing baselines, including 15 specialized models (or non-LLM baselines) and 7 LLM-based models to demonstrate its superiority in molecular property prediction. Specifically, SSLP[50] is a self-supervised learning platform to pretrain DL networks with over 700 million unlabeled SMILES data. GraphConv,[19] Weave[51] and SchNet[17] are models based on graph convolutional networks. MPNN[12] is the message passing model with edge features, and MGCN[13] is a variant of MPNN. GROVER[52] is a model based on the graph transformer. MolCLR[53] and GraphCL[54] are contrastive learning based methods. AttentiveFP[45] is the base model of the AttentiveFP module in our model. MvMRL[55] is a multiview molecular representation learning method, while GSL-MPP[56] is a model based on graph structure learning. GraSeq,[35] GraphMVP[38] and AGBT[41] are three multimodal models, where GraSeq and GraphMVP are models combining 1D and 2D molecular information, and AGBT combines 1D and 3D information. MoMu,[26] MoleculeSTM,[27] CaR,[57] LLaMA-2−7B chat,[23] InstructMol,[28] Graph2Token[43] and MolGraph-LarDo[32] are the 7 LLM-based models, among which MoMu, MoleculeSTM, InstructMol, Graph2Token and MolGraph-LarDo combine text representations with graph representations. The text encoder backbone of MoMu and MoleculeSTM is SciBERT.[58]

Experimental Setup. *Data Set Partitioning.* We use the scaffold splitting method[59] to divide each data set into training, validation, and test sets with a ratio of 8:1:1. Following previous works,[52,60] we report the average results and standard deviations of three independent runs using three different random seeds.

Evaluation Metrics. As recommended in MoleculeNet,[48] we use ROC-AUC to evaluate all classification tasks on data sets BBBP, BACE, ClinTox, SIDER, and Tox21. For regression tasks of data sets FreeSolv, ESOL and Lipo, we employ the RMSE metric to evaluate performance, while MAE is utilized for the task of QM7.

Implementation Details. We implement LLM-MPP using Pytorch and train the model with Adam optimizer. We run the experiments on GPU servers equipped with NVIDIA GeForce RTX 3090 and NVIDIA GeForce RTX 4090 with 24GB of memory. The settings of main hyperparameters are presented in Table 4. The training time of our model depends on the size and complexity of the data set used. For instance, training on the BACE data set takes approximately 380 s per epoch, while the BBBP data set requires around 500 s per epoch. Naturally, larger data sets entail more training time. However, as all the MoleculeNet data sets used in our experiments are relatively small in scale, the overall training process is feasible on standard GPU hardware. This ensures that our method is computationally practical for real-world applications.

Table 4. Hyperparameter Settings in LLM-MPP[a]

hyperparameter	value
early stop	15−30
max epochs	200
learning rate	10^{-5}
#cross attention heads	2
#GNN layers	4
GNN hidden dim	300
LoRA rank	4
LoRA alpha	16
prompt max length	800

[a]Note: the parameter for early stop is different from data set to data set.

■ RESULTS

Performance on Benchmark Data Sets. Tables 5 and 6 present a comprehensive performance comparison of our method with 19 baselines on the classification and regression benchmark data sets, respectively. We can see that our proposed model, LLM-MPP, achieves the best performance on four (BBBP, ClinTox, Tox21, SIDER) of the five classification data sets, and ranks first on two (FreeSolv, QM7) of the four regression data sets and second on the regression data set ESOL. In summary, LLM-MPP obtains six best results and one second

Table 5. Performance Comparison among Different Models on Various Data Sets for Classification Tasks[a]

data sets	BBBP	BACE	ClinTox	Tox21	SIDER
			non-LLM based baselines		
SSLP[50]	0.953 ± 0.009	0.872 ± 0.036	0.939 ± 0.047	–	–
GraphConv[19]	0.877 ± 0.036	0.854 ± 0.011	0.845 ± 0.051	0.772 ± 0.041	0.593 ± 0.035
Weave[51]	0.837 ± 0.065	0.791 ± 0.008	0.823 ± 0.023	0.741 ± 0.044	0.543 ± 0.034
SchNet[17]	0.847 ± 0.024	0.750 ± 0.033	0.717 ± 0.042	0.767 ± 0.025	0.545 ± 0.038
MPNN[12]	0.913 ± 0.041	0.815 ± 0.044	0.879 ± 0.054	0.808 ± 0.024	0.595 ± 0.030
MGCN[13]	0.850 ± 0.064	0.734 ± 0.030	0.634 ± 0.042	0.707 ± 0.016	0.552 ± 0.018
GROVER[52]	0.911	0.858	0.884	0.803	0.624
MolCLR[53]	0.738 ± 0.020	0.788 ± 0.050	0.867 ± 0.010	0.747 ± 0.080	0.669 ± 0.012
GraphCL[54]	0.696 ± 0.007	0.753 ± 0.014	0.759 ± 0.026	0.738 ± 0.007	0.605 ± 0.009
AttentiveFP[45]	0.908 ± 0.050	0.863 ± 0.015	0.933 ± 0.020	0.807 ± 0.020	0.605 ± 0.060
MvMRL[55]	0.963 ± 0.019	0.878 ± 0.336	0.976 ± 0.013	0.794 ± 0.014	0.607 ± 0.023
GSL-MPP[56]	0.957 ± 0.008	**0.896 ± 0.007**	0.947 ± 0.020	0.800 ± 0.021	0.652 ± 0.014
GraSeq[35]	0.926 ± 0.011	–	0.904 ± 0.004	0.755 ± 0.004	0.609 ± 0.009
GraphMVP[38]	0.724 ± 0.016	0.812 ± 0.009	0.791 ± 0.028	0.759 ± 0.005	0.639 ± 0.012
AGBT[41]	0.763	–	–	–	–
			LLM based baselines		
MoMu[26]	0.705 ± 0.020	0.771 ± 0.014	0.799 ± 0.041	0.756 ± 0.003	0.605 ± 0.009
MoleculeSTM[27]	0.708 ± 0.019	0.820 ± 0.004	0.925 ± 0.011	0.769 ± 0.005	0.637 ± 0.008
CaR[57]	0.820 ± 0.042	0.807 ± 0.014	0.842 ± 0.176	–	0.561 ± 0.018
LLaMA-2–7B chat[23]	0.656	0.748	–	–	–
Graph2Token[43]	0.735	0.850	–	0.792	–
InstructMol[28]	0.700	0.859	–	–	–
MolGraph-LarDo[32]	0.731 ± 0.015	0.831 ± 0.012	–	–	0.610 ± 0.023
LLM-MPP (ours)	**0.973 ± 0.005**	0.874 ± 0.016	**0.980 ± 0.004**	**0.818 ± 0.004**	**0.683 ± 0.015**

[a]The best results are highlighted in bold, while the second-best results are underlined. "–" means data unavailable.

Table 6. Performance Comparison among Different Models on Various Datasets for Regression Tasks[a]

data sets	FreeSolv	ESOL	Lipo	QM7
		non-LLM based baselines		
SSLP[50]	**0.891 ± 0.075**	0.915 ± 0.010	0.717 ± 0.029	–
GraphConv[19]	2.900 ± 0.135	1.068 ± 0.050	0.712 ± 0.049	118.875 ± 20.219
Weave[51]	2.398 ± 0.250	1.158 ± 0.055	0.813 ± 0.042	94.688 ± 2.705
SchNet[17]	3.215 ± 0.755	1.045 ± 0.064	0.909 ± 0.098	74.204 ± 4.983
MPNN[12]	2.185 ± 0.952	1.167 ± 0.430	0.672 ± 0.051	112.960 ± 17.211
MGCN[13]	3.349 ± 0.097	1.266 ± 0.147	1.113 ± 0.041	77.623 ± 4.734
GROVER[52]	1.987	0.911	0.643	89.408
MolCLR[53]	2.39 ± 0.14	1.16 ± 0.00	0.78 ± 0.01	–
GraphCL[54]	–	–	–	–
AttentiveFP[45]	2.030 ± 0.420	0.853 ± 0.060	0.650 ± 0.030	126.690 ± 4.020
MvMRL[55]	1.770 ± 0.229	0.856 ± 0.088	0.686 ± 0.037	–
GSL-MPP[56]	1.974 ± 0.315	**0.733 ± 0.071**	0.693 ± 0.063	60.47 ± 4.094
GraSeq[35]	2.218 ± 0.184	1.034 ± 0.035	0.780 ± 0.012	–
GraphMVP[38]	–	1.029	0.681	–
AGBT[41]	0.994	–	**0.570**	–
		LLM based baselines		
MoMu[26]	–	–	–	–
MoleculeSTM[27]	–	–	–	–
CaR[57]	–	0.960 ± 0.090	1.020 ± 0.060	–
LLaMA-2–7B chat[23]	–	–	–	–
Graph2Token[43]	–	–	–	–
InstructMol[28]	–	–	–	–
MolGraph-LarDo[32]	2.356 ± 0.079	1.277 ± 0.054	0.748 ± 0.017	74.910 ± 5.840
LLM-MPP (ours)	1.505 ± 0.232	0.829 ± 0.024	0.664 ± 0.006	**45.578 ± 1.796**

[a]The best results are highlighted in bold, and the second-best results are underlined. "–" means data unavailable.

best result on the 9 data sets in total, which demonstrates the effectiveness and superiority of our model in most cases.

Concretely, our model demonstrates outstanding performance in classification tasks, achieving a 1.4% improvement in ROC-AUC over the second-best MolCLR model on the SIDER

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Table 7. Ablation Study Results of LLM-MPP and Its Variants on Classification Data Sets

Variant Model	Module Selection				Classification (ROC-AUC)		
	LoRA	CL	CoT	Text	BACE	SIDER	ClinTox
Freeze LLM		✓	✓	✓	0.737 ± 0.004	0.596 ± 0.003	0.952 ± 0.009
w/o CL	✓		✓	✓	0.859 ± 0.024	0.662 ± 0.011	0.961 ± 0.011
w/o CoT	✓	✓		✓	0.737 ± 0.013	0.654 ± 0.009	0.977 ± 0.015
w/o Text	✓	✓	✓		0.855 ± 0.032	0.593 ± 0.007	0.974 ± 0.004
w/o CoT & Text	✓	✓			0.727 ± 0.057	0.604 ± 0.007	0.971 ± 0.005
LLM-MPP	✓	✓	✓	✓	$\mathbf{0.874 \pm 0.016}$	$\mathbf{0.683 \pm 0.015}$	$\mathbf{0.980 \pm 0.004}$

Table 8. Ablation Study Results of LLM-MPP and Its Variants on Regression Data Sets

Variant Model	Module Selection				Regression (RMSE)		
	LoRA	CL	CoT	Text	ESOL	FreeSolv	QM7
Freeze LLM		✓	✓	✓	0.870 ± 0.028	1.564 ± 0.097	54.171 ± 2.197
w/o CL	✓		✓	✓	0.954 ± 0.086	1.781 ± 0.321	46.225 ± 0.371
w/o CoT	✓	✓		✓	0.859 ± 0.024	2.016 ± 0.111	45.910 ± 1.673
w/o Text	✓	✓	✓		0.964 ± 0.036	1.862 ± 0.141	45.579 ± 1.125
w/o CoT & Text	✓	✓			1.409 ± 0.025	4.128 ± 0.345	49.805 ± 5.264
LLM-MPP	✓	✓	✓	✓	$\mathbf{0.829 \pm 0.024}$	$\mathbf{1.505 \pm 0.232}$	$\mathbf{45.578 \pm 1.796}$

data set. Our approach significantly outperforms the existing LLM-based models, and surpasses the specialized (non-LLM) models in most cases. On the BBBP, BACE, ClinTox, and Tox21 data sets, our model outperforms the second-best LLM-based models by 15.3, 1.5, 5.5, and 2.6%, respectively, showcasing its effective utilization of LLM's encoding capability for molecular representation and fusion.

As for regression tasks, our model still achieves the best results on the FreeSolv and QM7 data sets. Notably, it reduces the RMSE metric by 14.89 compared to the second-best model GSL-MPP on the QM7 data set. The slightly lower performance in regression tasks than in classification tasks may be due to the limitations of large language models in regression tasks. However, the graph representations generated by the graph neural network in our multimodal fusion method effectively compensate for the shortcomings of LLMs.

Ablation Studies. *Effects of Different Modules.* To investigate the effect of each component of our method, an ablation study is conducted on six data sets, including three classification data sets (BACE, SIDER, ClinTox) and three regression data sets (ESOL, FreeSolv, QM7). We consider five variant models for comparison as follows:

- Freeze LLM: this variant does not use LoRA fine-tuning, freezing all the parameters in the LLM.
- w/o CL: this variant does not use contrastive learning, i.e., the final loss function does not contain the contrastive loss.
- w/o CoT: this variant excludes chain-of-thought prompting, relying only on 1D SMILES strings and textual descriptions within the prompt for the LLM.
- w/o Text: this variant excludes textual description data, relying only on 1D SMILES strings and chain-of-thought text within the prompt for the LLM.
- w/o CoT & Text: this variant excludes chain-of-thought prompting and textual description data, relying only on 1D SMILES strings within the prompt for the LLM.

Tables 7 and 8 present the results of these variants and LLM-MPP on classification and regression data sets, respectively. From the results in the two tables, we can see that removing any module results in degraded performance, while the complete model consistently achieves the best result on all the six data sets. However, the effect of each module varies with data set. For example, on the BACE data set, the chain-of-thought module is particularly important — removing it leads to a significant performance drop. On the SIDER data set, excluding the text description has a more obvious negative effect on predictive accuracy. While on the ClinTox data set, the impact of removing these modules is trivial. Even using only SMILES strings as prompt achieves good prediction results. In the regression tasks, excluding CoT and textual description significantly affects performance, likely because LLMs tend to perform less effectively in regression tasks than in classification tasks, requiring additional fine-tuning techniques to boost predictive accuracy.

To better understand the varying contributions of different modules in our framework across data sets, we analyze this phenomenon from two complementary chemical perspectives: (1) *The task nature and complexity*, which reflects the underlying biochemical or physical mechanisms involved in the prediction target. The BACE task involves nuanced structure−activity relationships, including stereochemistry and binding pocket compatibility. Therefore, CoT allows the model to reason about intermediate biochemical interactions, which are essential in this binary classification task where small structural changes can dramatically impact binding. The FreeSolv task also shows strong reliance on CoT. The thermodynamic process it models requires subtle physicochemical reasoning that goes beyond surface-level features. The CoT module likely enables the LLM to approximate concepts like entropy or solvation shells, contributing to better generalization. (2) *The data characteristics*, such as feature-label correlation, data set size, chemical diversity, and class distribution. ClinTox exhibits only small performance differences across ablations, which is partly due to the presence of well-defined structural alerts and toxicophores, which are relatively easy to detect using SMILES or graph features alone. In addition, high class imbalance may overshadow subtle performance differences. Sider's reliance on textual description is likely because many side effects are tied to specific pharmacological classes or mechanistic pathways that are not directly inferable from SMILES strings and molecular graphs alone but may be related to textual descriptions. Solubility depends heavily on hydrophilicity, hydrogen bonding potential, and polarity, which are often described explicitly in curated textual descriptions, aiding the LLM's interpretation, which makes ESOL's performance rely heavily on textual description. QM7 exhibits the largest drop in performance when the LLM is

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Table 9. Results of Different Values of η_t and η_g on Different Datasets

		classification (ROC-AUC)		regression (RMSE)	
η_t	η_g	BACE	SIDER	ESOL	FreeSolv
0.0	0.8	0.854 ± 0.010	0.654 ± 0.017	0.845 ± 0.023	**1.505 ± 0.232**
0.0	0.5	0.848 ± 0.011	0.667 ± 0.014	0.917 ± 0.001	1.961 ± 0.231
0.3	0.3	0.851 ± 0.017	0.679 ± 0.014	0.958 ± 0.015	1.877 ± 0.024
0.5	0.0	0.874 ± **0.016**	**0.683 ± 0.015**	0.883 ± 0.028	1.968 ± 0.189
0.8	0.0	0.865 ± 0.013	0.676 ± 0.011	0.916 ± 0.051	1.907 ± 0.274
0.0	0.0	0.850 ± 0.020	0.678 ± 0.004	0.899 ± 0.058	1.907 ± 0.274
0.5	0.5	0.850 ± 0.004	0.668 ± 0.012	**0.829 ± 0.024**	1.897 ± 0.235

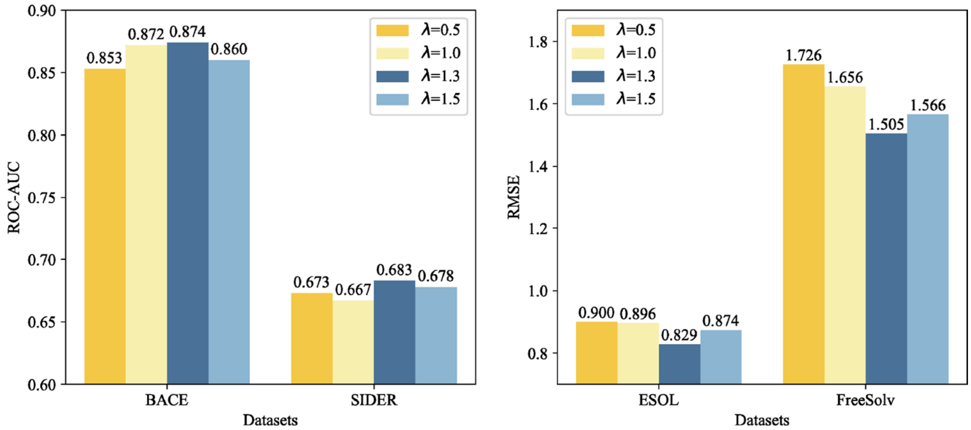

Figure 6. Results of different values of λ on different data sets. Here, the values of η_t and η_g are set to those with the best results on each data set.

frozen. This suggests that the LLM is required to adapt and fine-tune to quantum-specific patterns and physical properties during training. Freezing the LLM restricts this adaptability, harming performance.

As an example, we analyze the molecule (2r)-3-[2-Amino-6-(2-Methylphenyl)quinolin-3-Yl]-N-[(4r)-2,2-Dimethyltetrahydro-2h-Pyran-4-Yl]-2-Methylpropanamide from the BACE data set using the CoT-enabled LLM-MPP model. BACE is a key target in Alzheimer's disease therapy, where inhibitors reduce β-amyloid plaque formation. First, the model evaluates basic drug-like properties: the molecular weight (431.6 g/mol) meets oral bioavailability requirements (<500 g/mol); The XLogP3-AA (4.9) is slightly high (where the ideal range is <4) but can be optimized by adjusting polar groups to balance permeability and solubility; The hydrogen bond donor (2) and acceptor (4) counts conform to Lipinski's Rule of Five; The TPSA (77.2 $Å^2$) suggests moderate polarity, potentially affecting membrane permeability. Second, structural analysis identifies key pharmacophores: a rigid epoxyhexane ring (O1CCC···CC1(C)C) enhancing metabolic stability; The amide group (NC = O) acting as a hydrogen bond donor and potentially forming a directional interaction with the Asp residue in the catalytic site of BACE; The polycyclic aromatic systems (benzene ring, quinoline core) filling the target hydrophobic pocket through hydrophobic interaction and π-π stacking, and two stereocenters (2R, 4R) improving spatial complementarity with the target. Finally, LLM-MPP integrates multidimensional data through CoT and infers that the molecule possesses the typical characteristics of a BACE inhibitor: moderate molecular weight, rigid skeleton, and key functional groups that could support target binding, but high LogP and TPSA may limit its efficacy. The CoT-driven LLM-MPP emulates the logical analytical processes employed by medicinal chemists by implementing a modular reasoning framework comprising target recognition, physicochemical screening, structural analysis, and comprehensive evaluation. This approach enhances both the robustness and interpretability of molecular property predictions.

On all the data sets, removing the LoRA fine-tuning module leads to performance degradation, highlighting the importance of task-specific fine-tuning in adapting LLMs for specific applications, except for on the FreeSolv data set, where the impact of LoRA is relatively obscure. LoRA enables LLMs to leverage their pretrained knowledge more effectively within domain-specific contexts, yielding more accurate representations and better prediction performance. This suggests that fine-tuning with LoRA improves the model's ability to focus on relevant information on specific data sets, thereby making it more accurate in these specific tasks. Removing the contrastive learning module affects the predictive performance on each data set, though the impact is relatively minor on the whole, indicating that while contrastive learning can enhance modality alignment and aid feature learning across different modalities, our text representation and graph representation learning modules are already highly effective in capturing meaningful features independently.

Effects of Hyperparameters. Here, we conduct extensive experiments to check the effects of major hyperparameters on four data sets: two classification data sets (BACE and SIDER) and two regression data sets (ESOL and FreeSolv).

Table 9 presents the results for different values of η_t and η_g, which balance the contributions of text representations, graph representations, and the representations output from the cross-attention module for fusion. On the BACE and SIDER data sets, the best results are achieved when the values of η_t and η_g are set to 0.5 and 0.0, respectively. This indicates that a slightly higher ratio of text representations during fusion can enhance the performance of classification tasks, leveraging the remarkable information extraction capabilities of LLMs. However, on the

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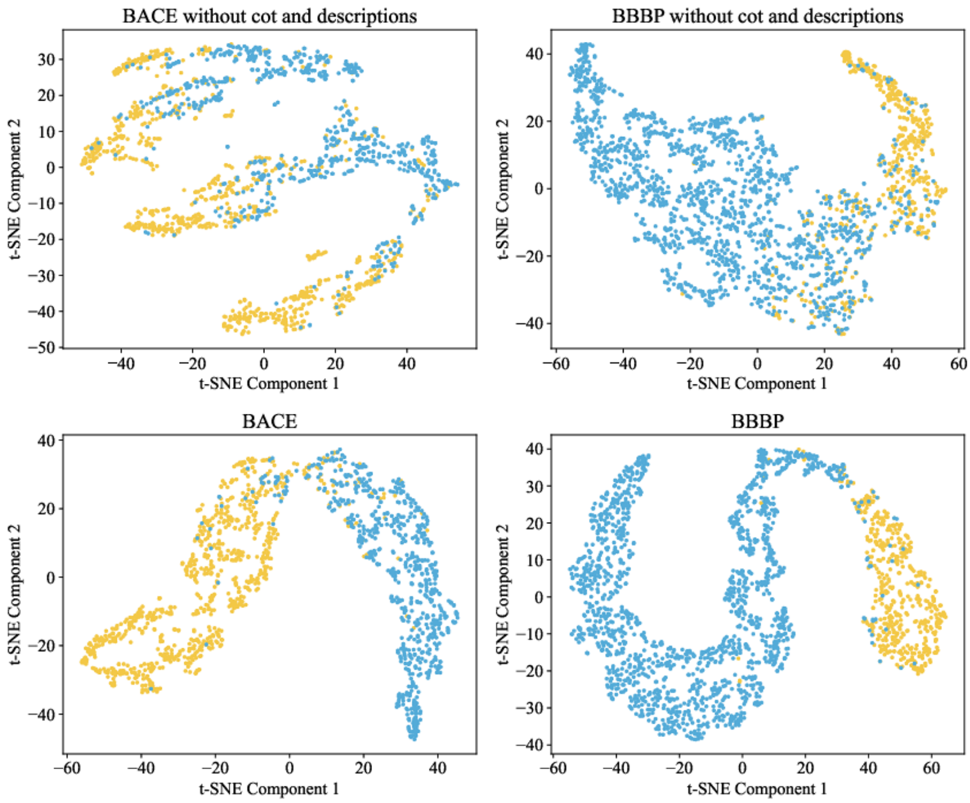

Figure 7. Visualization of molecular representations learned by LLM-MPP (bottom) and the variant without chain-of-thought technique and textual descriptions (top) on BACE and BBBP data sets via t-SNE. Yellow points and blue points represent molecular representations with label 0 and 1, respectively.

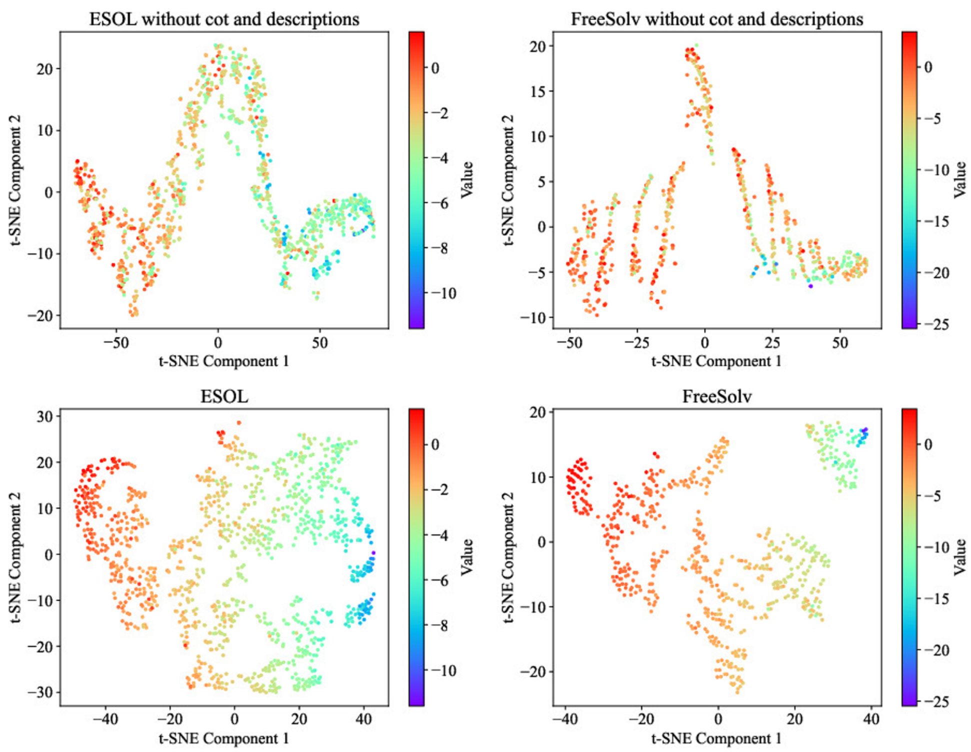

Figure 8. Visualization of molecular representations learned by LLM-MPP (bottom) and the variant without chain-of-thought technique and textual descriptions (top) on the ESOL and FreeSolv data sets via t-SNE. The colors of the points change from red to purpose as the property value decreases.

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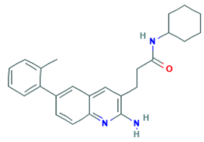
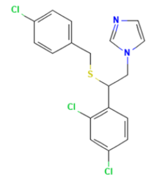
Dataset	BACE	ClinTox
2D Structure		
SMILES	O=C(NC1CCCCC1)CCc1cc2cc(ccc2nc1N)-c1ccccc1C	c1cc(ccc1CSC(Cn2ccnc2)c3ccc(cc3Cl)Cl)Cl
Molecular Formula	$C_{25}H_{29}N_3O$	$C_{18}H_{15}Cl_3N_2S$
CoT Text	To predict BACE-1 inhibition of this molecule, we'll need to analyze its structure and properties. Here's a step-by-step approach: **1. Molecular structure analysis:** The molecule has a complex structure with multiple rings, including a cycloalkyl group, a phenyl ring, and a pyrroline ring. This suggests that it may have some flexibility and ability to adopt different conformations. **2. Binding site analysis:** Bace-1 is a metalloenzyme that cleaves the β-secretase site of amyloid precursor protein (APP). The active site is relatively small, with a binding pocket that accommodates a small molecule with specific interactions. The binding site has been well-studied, so we can use this information to guide our analysis. In summary, based on this analysis and prediction, this novel compound is likely to be a BACE-inhibitor, although further experimental validation is necessary to confirm its activity.	The structure of the molecule is a azole, and this class of compounds has been widely used as antifungal agents. The molecule has a structure similar to ketoconazole, which is a well-known antifungal agent. The predicted logP value is 6.1, which is relatively high, indicating that the molecule may have poor aqueous solubility. However, the solubility of the molecule is described as "Slightly soluble", which may be due to the presence of the sulfur atom, which can improve the solubility of the molecule. Based on the above analysis, it is predicted that the molecule may have antifungal activity, and the toxicity of the molecule may be relatively low due to its similarity to ketoconazole. However, further studies are needed to confirm the efficacy and safety of the molecule.
Predicted Label	Yes	No

Figure 9. Texts generated by chain-of-thought prompting for two molecules from the BACE and ClinTox data sets, respectively.

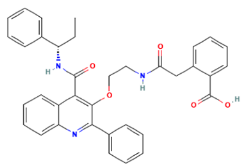
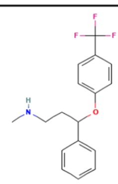
Dataset	BBBP	BBBP
2D Structure		
SMILES	c1ccc2c(c(C(N[C@@H](CC)c3ccccc3)=O)c(OCCNC (Cc3c(cccc3)C(=O)O)=O)c(c3ccccc3)n2)c1	C2=C(C(OC1=CC=C(C=C1)C(F)(F)F)CCNC)C=CC=C2
Molecular Formula	$C_{36}H_{33}N_3O_5$	$C_{17}H_{18}F_3NO$
CoT Text	Here's a step-by-step analysis: **1. Molecular weight**: The molecular weight of your molecule is approximately 588 Da, which is relatively high. Higher molecular weights can make it more difficult for a compound to penetrate the brain. **2. Rotating bonds:** The presence (12) of rotating bonds in your compound may indicate flexibility, allowing it to adapt to the lipid bilayer of cell membranes. Flexibility can facilitate membrane permeation. **3. Charge:** As your molecular structure is neutral (Formally charged), it is less affected by electrostatic interactions, but it can still interact with charged molecules in blood and brain tissue. **4. Polar surface area:** With a topological polar surface of approximately118 Å2, your substance has some polar regions that can interact favorably with water and other polar molecules. While this might facilitate solubility, it could also hinder BBB penetration. **5. Atom count:** Heavy atoms (44) in this molecule may contribute to its overall size and molecular mass, potentially affecting its ability to traverse the cell membrane. Considering these factors, I would predict that your compound will have limited Brain Blood Barrier penetration. The high molecular weight, polar surface area, atom count may hinder its passage across this barrier.	1. The molecular weight of the molecule is 309.33, which is less than 500, so it is likely to cross the blood-brain barrier. 2. The molecule has 5 hydrogen bond acceptors, which is less than 5, so it is likely to cross the blood-brain barrier. 3. The molecule has 6 rotatable bonds, which is less than 10, so it is likely to cross the blood-brain barrier. 4. The LogP value of the molecule is 4, which is less than 5, so it is likely to cross the blood-brain barrier. 5. The polar surface area of the molecule is 21.3, which is less than 120, so it is likely to cross the blood-brain barrier. Based on these rules, we can conclude that the molecule is likely to cross the blood-brain barrier.
Predicted Label	No	Yes

Figure 10. Texts generated by chain-of-thought prompting for two molecules from the BBBP data set.

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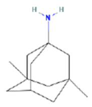
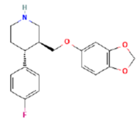
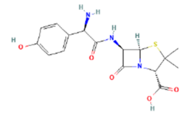
Drug Name	Memantine	Paroxetine	Diazepam	Amoxicillin
2D Structure				
SMILES	CC12CC3CC(C1)(CC(C3)(C2)N)C	C1CNC[C@H]([C@@H]1C2=CC=C(C=C2)F)COC3=CC4=C(C=C3)OCO4	CN1C(=O)CN=C(C2=C1C=CC(=C2)Cl)C3=CC=CC=C3	CC1([C@@H](N2[C@H](S1)[C@@H](C2=O)NC(=O)[C@@H](C3=CC=C(C=C3)O)N)C(=O)O)C
Drug Overview	Memantine, sold under the brand name Namenda among others, is a medication used to slow the progression of moderate-to-severe Alzheimer's disease. It is taken by mouth.	Paroxetine, sold under the brand name Paxil among others, is an antidepressant of the selective serotonin reuptake inhibitor (SSRI) class. It is used to treat major depressive disorder, obsessive-compulsive disorder, panic disorder, social anxiety disorder, post-traumatic stress disorder, generalized anxiety disorder, and premenstrual dysphoric disorder. It has also been used in the treatment of premature ejaculation and hot flashes due to menopause.	Diazepam, sold under the brand name Valium among others, is a medicine of the benzodiazepine family that acts as an anxiolytic. It is used to treat a range of conditions, including anxiety, seizures, alcohol withdrawal syndrome, muscle spasms, insomnia, and restless legs syndrome. It may also be used to cause memory loss during certain medical procedures.	Amoxicillin is an antibiotic medication belonging to the aminopenicillin class of the penicillin family. The drug is used to treat bacterial infections such as middle ear infection, strep throat, pneumonia, skin infections, odontogenic infections, and urinary tract infections.
Blood Brain Barrier	Yes	Yes	Yes	No
Predicted Probability	0.989	0.569	0.801	0.117
Predicted Label	Yes	Yes	Yes	No

Figure 11. Case study on blood-brain barrier penetration prediction for four drug molecules. The *Blood Brain Barrier* row indicates the true penetration capabilities of the drugs. The *Predicted Probability* row presents the probabilities output by our model, where values close to 1 suggest a high likelihood of BBB penetration, and values close to 0 indicate a low likelihood. The *Predicted Label* row shows the binary labels predicted by our model, indicating whether the drug is expected to penetrate the blood-brain barrier. The *Drug Overview* is sourced from Wikipedia.

FreeSolv data set, increasing the weight of graph representations improves the accuracy of property predictions, owing to the AttentiveFP model's power in regression tasks.

Figure 6 illustrates the results for different values of λ, which balances the contributions of predictive loss and contrastive loss in the final loss function. The results indicate that for both classification and regression tasks, the model performs best when λ is set to 1.3, suggesting that a relatively larger value of λ promotes the alignment of different modalities, which ensures better representation consistency of different modalities, allowing the model to learn more accurate and effective molecular representations.

Visualization of Molecular Representations. We apply t-SNE[61] to visualize molecular representations from the BACE, BBBP, ESOL, and FreeSolv data sets produced by two model variations: (1) our model, which includes both the chain-of-thought technique and textual descriptions for multimodal integration, and (2) a simplified model without the chain-of-thought reasoning and textual descriptions. t-SNE is a dimensionality reduction technique that projects high-dimensional data into a lower-dimensional space (here 2D) to reveal the data's underlying structure. By comparing the t-SNE plots of the two models, we can visually assess the impact of the two components on the quality of the molecular representations. Figures 7 and 8 illustrate the representations of classification data sets (BACE and BBBP) and regression data sets (ESOL and FreeSolv), respectively.

On the two classification data sets, the molecular representations generated by our model exhibit clear cluster structures according to their labels, whereas the model without chain-of-thought and textual descriptions produces representations with less distinct boundaries, especially on the BACE data set. This is consistent with the findings from our previous ablation studies. On the regression data sets, the representations from our model also show clear distribution patterns: on the ESOL data set, the values of molecular properties decrease from left to right in the figure, and on the FreeSolv data set, the property values decrease from the lower left to the upper right in the figure. In contrast, the distribution patterns are less pronounced with the simplified model, demonstrating the effectiveness of the proposed modules in our model.

Interpretability Analysis. Here, we examine the texts generated by using the chain-of-thought technique for molecular property prediction, which describe the reasoning process and demonstrate the interpretability of the model in MPP.

Figure 9 presents the CoT texts of two specific molecules from the BACE and ClinTox data sets respectively, and Figure 10 shows the CoT texts of two molecules from the BBBP data set. The chain-of-thought reasoning process reveals how the model understands and utilizes the multimodal information provided in the data sets to infer molecular properties. By explicitly showcasing the model's reasoning steps in the CoT text, we can trace how different molecular features and information contribute to the prediction, thus enhancing the interpretability of the model's prediction.

For instance, consider the prediction of $C_{36}H_{33}N_3O_5$ from the BBBP data set in Figure 10, the CoT text describes how the LLaMA model uses and analyzes the textual description of the molecule. Specifically, the model considers molecular weight and heavy atom count, notices that a higher molecular weight (approximately 588 Da) and a large number of heavy atom count (44 for this molecule) may reduce the likelihood of blood-brain barrier penetration, because larger molecules are more difficult to penetrate. Furthermore, the model considers rotating bonds, recognizes that the presence of 12 rotatable bonds enhances molecular flexibility, which may facilitate adaptation to the lipid bilayer of cell membranes, aiding permeability. It also assesses charge, regards that the molecule is neutral (formally uncharged) but can still interact with charged molecules in the blood and brain tissue. Lastly, the model evaluates the molecule's polar surface area, identifies that a topological polar surface area of approximately 118 Å^2 provides polar regions that interact favorably with water and other polar molecules, which may hinder penetration through the blood-brain barrier. In summary, the CoT text demonstrates how the model assesses the likelihood of the molecule crossing the blood-brain barrier

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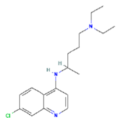
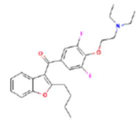
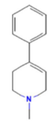
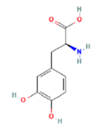
Drug Name	Chloroquine	Amiodarone	MPTP	Levodopa
2D Structure				
SMILES	CCN(CC)CCCC(C)NC1=C2C=CC(=CC2=NC=C1)Cl	CCCCC1=C(C2=CC=CC=C2O1)C(=O)C3=CC(=C(C(=C3)I)OCCN(CC)CC)I	CN1CCC(=CC1)C2=CC=CC=C2	C1=CC(=C(C=C1C[C@@H](C(=O)O)N)O)O
Drug Overview	Chloroquine is an antiparasitic medication that treats malaria. It works by increasing the levels of haeme in the blood, a substance toxic to the malarial parasite. This kills the parasite and stops the infection from spreading. Certain types of malaria, resistant strains, and complicated cases typically require different or additional medication.	Amiodarone is an antiarrhythmic medication used to treat and prevent a number of types of cardiac dysrhythmias. This includes ventricular tachycardia, ventricular fibrillation, and wide complex tachycardia, atrial fibrillation, paroxysmal supraventricular tachycardia. Evidence in cardiac arrest, however, is poor. It can be given by mouth, intravenously, or intraosseously.	MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine) is an organic compound. It is classified as a tetrahydropyridine. It is of interest as a precursor to the monoaminergic neurotoxin MPP+, which causes permanent symptoms of Parkinson's disease by destroying dopaminergic neurons in the substantia nigra of the brain. It has been used to study disease models in various animals.	Levodopa, also known as L-DOPA and sold under many brand names, is a dopaminergic medication which is used in the treatment of Parkinson's disease and certain other conditions like dopamine-responsive dystonia and restless legs syndrome. The drug is usually used and formulated in combination with a peripherally selective aromatic L-amino acid decarboxylase (AAAD) inhibitor like carbidopa or benserazide.
Clinical Trial Toxicity	Yes	Yes	Yes	No
Predicted Probability	0.880	0.930	0.899	0.229
Predicted Label	Yes	Yes	Yes	No

Figure 12. Case study on clinical trial toxic prediction for four drug molecules. The *Clinical Trial Toxicity* row indicates the true toxicity of the drugs. The *Predicted Probability* row presents the probabilities output by our model, where values close to 1 suggest a high likelihood of clinical trial toxicity, and values close to 0 indicate a low likelihood. The *Predicted Label* row shows the binary labels predicted by our model, indicating whether the drug is expected to have clinical trial toxicity. The *Drug Overview* is sourced from Wikipedia.

by utilizing the available information, which gives a kind of interpretation of the model's prediction.

Case Study. To further assess the performance of LLM-MPP, we use it to predict the properties of several FDA-approved drugs. Figures 11 and 12 show the case studies on blood-brain barrier penetration prediction and clinical trial toxicity prediction.

We select memantine, paroxetine, diazepam, and amoxicillin for the blood-brain barrier penetration prediction task. Memantine[1] is an NMDA (*N*-methyl-D-aspartate) receptor antagonist, which is primarily used to treat moderate to severe Alzheimer's disease and is known for its ability to cross the BBB effectively. Paroxetine[2], a selective serotonin reuptake inhibitor, is commonly prescribed for managing depression and anxiety disorders, leveraging its capacity to penetrate the BBB to exert therapeutic effects. Diazepam[3] is a benzodiazepine used for anxiety, seizures, and muscle spasms, and it is also BBB-permeable. In contrast, amoxicillin[4], a beta-lactam antibiotic frequently used to treat bacterial infections, demonstrates poor BBB penetration, limiting its efficacy against infections localized in the central nervous system. Our model accurately predicts the BBB penetration property labels for these four drugs, assigning high probabilities to memantine, paroxetine, and diazepam, and a low probability to amoxicillin.

In the clinical trial toxicity prediction task, we choose chloroquine, amiodarone, MPTP and levodopa for the study. Chloroquine[5], an antimalarial drug approved by the FDA in 1949, has been associated with severe toxic effects at high doses, including cardiotoxicity and retinopathy, which limit its clinical use. Its emergency use authorization (EUA) for COVID-19 treatment was revoked by the FDA in 2020 due to insufficient evidence of efficacy and safety concerns. Additionally, severe toxicity cases, such as the fatal outcome of a 42-year-old man who received a 250 mg intravenous dose, underscore the danger associated with its misuse. Amiodarone[6], an antiarrhythmic drug, is effective in treating life-threatening ventricular arrhythmias but carries a high risk of severe organ toxicities and potentially increased mortality.[62] MPTP (1-Methyl-4-phenyl-1,2,3,6-tetrahydropyridine)[7] is a neurotoxin, which can cause permanent Parkinsonian symptoms by selectively destroying dopaminergic neurons in the substantia nigra. In contrast, levodopa[8], widely used for the treatment of Parkinson's disease, exhibits relatively low clinical toxicity.[63] Though it can cause dose-dependent side effects such as nausea and dyskinesias with prolonged use, these issues are generally manageable and far less severe than the toxicities associated with the other drugs in this study. Our LLM-MPP successfully predicts the clinical toxicity of these drugs, demonstrating its power of effective prediction in real-world scenarios.

■ CONCLUSIONS

In this paper, we propose a novel effective and interpretable multimodal molecular property prediction method based on large language models. Our method, LLM-MPP, leverages LLaMA3 to extract and align features from SMILES strings and molecular text descriptions. For molecular graph representation, it utilizes the AttentiveFP GNN network. It combines these representations using a cross-attention mechanism along with a weighted sum of the textual representation, graph representation, and cross-attention output to obtain the final molecular representation, and utilizes a contrastive loss to align representations from different modalities. An MLP network is then employed for the final property prediction.

In summary, our method addresses the alignment challenges between SMILES strings and natural language textual descriptions related to molecule properties by utilizing the extensive prior knowledge and powerful encoding capabilities of large language models, resulting in rich text representations. The cross-attention mechanism and weighted sum integration effectively balance and fuse text and graph representations, overcoming the limitations of large language models in regression tasks for molecular property prediction. We introduce the chain-of-thought technique to improve the interpretability and transparency of predictions. Our model achieves state-of-

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the-art performance on five out of the nine MoleculeNet data sets, which is better than previous non-LLM baselines and LLM-based approaches. We further validate the effectiveness of our model and its individual components through ablation studies, hyper-parameter tuning, and visualization of molecular representations, as well as case study.

However, our model does not achieve the best results on all data sets, and there is still room for improvement, particularly in regression tasks. In the future, we aim to refine our model further and develop innovative approaches to enhance its performance on these regression data sets. Additionally, given that few-shot scenarios are common in molecular property prediction for drug discovery, where limited data often leads to lower prediction accuracy, we plan to explore how to leverage the strong generalization capabilities of large language models to improve prediction accuracy in few-shot settings.

■ ASSOCIATED CONTENT

Data Availability Statement

The data and code supporting the findings of this study are available in the following GitHub repository: https://github.com/jinchang1223/LLM-MPP. The repository includes the implementation of our proposed model, preprocessing scripts, and instructions for reproducing the experiments. Any additional data sets used in this study are either publicly available from their respective sources or provided in the repository with appropriate references.

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Author Contributions

C.J., S.G. and S.Z. designed the research and experiments. C.J. conducted the experiments. C.J., S.G., S.Z. and J.G. wrote and reviewed the manuscript. S.Z. and J.G. supervised this research.

Notes

The authors declare no competing financial interest.

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■ ADDITIONAL NOTES

[1]https://pubchem.ncbi.nlm.nih.gov/compound/4054
[2]https://pubchem.ncbi.nlm.nih.gov/compound/43815
[3]https://pubchem.ncbi.nlm.nih.gov/compound/3016
[4]https://pubchem.ncbi.nlm.nih.gov/compound/33613
[5]https://pubchem.ncbi.nlm.nih.gov/compound/2719
[6]https://pubchem.ncbi.nlm.nih.gov/compound/2157
[7]https://pubchem.ncbi.nlm.nih.gov/compound/1388
[8]https://pubchem.ncbi.nlm.nih.gov/compound/6047

■ REFERENCES

(1) Liyaqat, T.; Ahmad, T.; Saxena, C. *Advancements in Molecular Property Prediction: A Survey of Single and Multimodal Approaches*, 2024; https://arxiv.org/abs/2408.09461.

(2) Ferreira, L. L.; Andricopulo, A. D. ADMET modeling approaches in drug discovery. *Drug discovery today* **2019**, *24*, 1157−1165.

(3) Dong, J.; Wang, N.-N.; Yao, Z.-J.; Zhang, L.; Cheng, Y.; Ouyang, D.; Lu, A.-P.; Cao, D.-S. ADMETlab: a platform for systematic ADMET evaluation based on a comprehensively collected ADMET database. *J. Cheminf.* **2018**, *10*, 29.

(4) Shen, J.; Nicolaou, C. A. Molecular property prediction: recent trends in the era of artificial intelligence. *Drug Discovery Today: Technologies* **2019**, *32*, 29−36.

(5) Wieder, O.; Kohlbacher, S.; Kuenemann, M.; Garon, A.; Ducrot, P.; Seidel, T.; Langer, T. A compact review of molecular property prediction with graph neural networks. *Drug Discovery Today: Technologies* **2020**, *37*, 1−12.

(6) Wu, Y.; Ni, X.; Wang, Z.; Feng, W. Enhancing drug property prediction with dual-channel transfer learning based on molecular fragment. *BMC Bioinf.* **2023**, *24*, 293.

(7) Weininger, D. SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules. *Journal of chemical information and computer sciences* **1988**, *28*, 31−36.

(8) Kuhn, H.; Braslavsky, S.; Schmidt, R. Chemical actinometry (IUPAC technical report). *Pure Appl. Chem.* **2004**, *76*, 2105−2146.

(9) Hochreiter, S. *Long Short-term Memory*; Neural Computation MIT-Press, 1997.

(10) Vaswani, A. Attention is all you need. In *Advances in Neural Information Processing Systems*, **2017**.

(11) Devlin, J. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* **2018**, No. 1810.04805.

(12) Gilmer, J.; Schoenholz, S. S.; Riley, P. F.; Vinyals, O.; Dahl, G. E. Neural message passing for quantum chemistry. *Int. Conf. Mach. Learn.* **2017**, 1263−1272.

(13) Lu, C.; Liu, Q.; Wang, C.; Huang, Z.; Lin, P.; He, L. Molecular property prediction: A multilevel quantum interactions modeling perspective. *Proceedings of the AAAI conference on artificial intelligence* **2019**, *33*, 1052−1060.

(14) Kipf, T. N.; Welling, M. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907* **2016**, No. 1609.02907.

(15) Veličković, P.; Cucurull, G.; Casanova, A.; Romero, A.; Lio, P.; Bengio, Y. Graph attention networks. *arXiv preprint arXiv:1710.10903* **2017**, No. 1710.10903.

(16) Hamilton, W.; Ying, Z.; Leskovec, J. Inductive representation learning on large graphs. *Adv. Neural Inf. Process. Syst.* **2017**, *30*, 1025−1035.

(17) Schütt, K.; Kindermans, P.-J.; Sauceda Felix, H. E.; Chmiela, S.; Tkatchenko, A.; Müller, K.-R. Schnet: A continuous-filter convolutional neural network for modeling quantum interactions. *Adv. Neural Inf. Process. Syst.* **2017**, *30*, 992−1002.

(18) Gasteiger, J.; Groß, J.; Günnemann, S. Directional message passing for molecular graphs. *arXiv preprint arXiv:2003.03123* **2020**, No. 2003.03123.

(19) Duvenaud, D. K.; Maclaurin, D.; Iparraguirre, J.; Bombarell, R.; Hirzel, T.; Aspuru-Guzik, A.; Adams, R. P. Convolutional networks on graphs for learning molecular fingerprints. *Adv. Neural Inf. Process. Syst.* **2015**, *28*, 5.

https://doi.org/10.1021/acs.jcim.5c00577
J. Chem. Inf. Model. 2025, 65, 5438−5455

(20) Todeschini, R.; Consonni, V. *Handbook of molecular descriptors*; John Wiley & Sons, 2008.
(21) Radford, A.; Wu, J.; Child, R.; Luan, D.; Amodei, D.; Sutskever, I.; et al. Language models are unsupervised multitask learners. *OpenAI Blog* **2019**, *1*, 9.
(22) Brown, T. B. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165* **2020**, No. 2005.14165.
(23) Touvron, H.; Lavril, T.; Izacard, G.; Martinet, X.; Lachaux, M.-A.; Lacroix, T.; Rozière, B.; Goyal, N.; Hambro, E.; Azhar, F.; et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971* **2023**, No. 2302.13971.
(24) DeepSeek-AI, D. G.; Yang, D.; Zhang, H.; Song, J.; Zhang, R.; Xu, R.; Zhu, Q.; Ma, S.; Wang, P.; Bi, X.; et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948* **2025**, No. 2501.12948.
(25) Wang, A. Glue: A multi-task benchmark and analysis platform for natural language understanding. *arXiv preprint arXiv:1804.07461* **2018**, No. 1804.07461.
(26) Su, B.; Du, D.; Yang, Z.; Zhou, Y.; Li, J.; Rao, A.; Sun, H.; Lu, Z.; Wen, J.-R. A molecular multimodal foundation model associating molecule graphs with natural language. *arXiv preprint arXiv:2209.05481* **2022**, No. 2209.05481.
(27) Liu, S.; Nie, W.; Wang, C.; Lu, J.; Qiao, Z.; Liu, L.; Tang, J.; Xiao, C.; Anandkumar, A. Multi-modal molecule structure−text model for text-based retrieval and editing. *Nature Machine Intelligence* **2023**, *5*, 1447−1457.
(28) Cao, H.; Liu, Z.; Lu, X.; Yao, Y.; Li, Y. Instructmol: Multi-modal integration for building a versatile and reliable molecular assistant in drug discovery. *arXiv preprint arXiv:2311.16208* **2023**, No. 2311.16208.
(29) Li, S.; Liu, Z.; Luo, Y.; Wang, X.; He, X.; Kawaguchi, K.; Chua, T.-S.; Tian, Q. Towards 3d molecule-text interpretation in language models. *arXiv preprint arXiv:2401.13923* **2024**, No. 2401.13923.
(30) Guo, T.; Nan, B.; Liang, Z.; Guo, Z.; Chawla, N.; Wiest, O.; Zhang, X.; et al. What can large language models do in chemistry? a comprehensive benchmark on eight tasks. *Adv. Neural Inf. Process. Syst.* **2023**, *36*, 59662−59688.
(31) Zhang, X.-C.; Wu, C.-K.; Yi, J.-C.; Zeng, X.-X.; Yang, C.-Q.; Lu, A.-P.; Hou, T.-J.; Cao, D.-S. Pushing the boundaries of molecular property prediction for drug discovery with multitask learning BERT enhanced by SMILES enumeration. *Research* **2022**, *2022*, No. 0004.
(32) Zhang, T.; Ren, Y.; Hou, C.; Lv, H.; Zhang, X. Molecular graph representation learning integrating large language models with domain-specific small models. *IEEE Int. Conf. Bioinf. Biomed.* **2024**, *2024*, 2928−2935.
(33) Fang, Y.; Liang, X.; Zhang, N.; Liu, K.; Huang, R.; Chen, Z.; Fan, X.; Chen, H. Mol-instructions: A large-scale biomolecular instruction dataset for large language models. *arXiv preprint arXiv:2306.08018* **2023**, No. 2306.08018.
(34) Wei, J.; Wang, X.; Schuurmans, D.; Bosma, M.; Xia, F.; Chi, E.; Le, Q. V.; Zhou, D.; et al. Chain-of-thought prompting elicits reasoning in large language models. *Adv. Neural Inf. Process. Syst.* **2022**, *35*, 24824−24837.
(35) Guo, Z.; Yu, W.; Zhang, C.; Jiang, M.; Chawla, N. V. Graseq: graph and sequence fusion learning for molecular property prediction. In *Proceedings of the 29th ACM international conference on information & knowledge management*, 2020; pp 435−443.
(36) Xie, A.; Zhang, Z.; Guan, J.; Zhou, S. Self-supervised learning with chemistry-aware fragmentation for effective molecular property prediction. *Briefings Bioinf.* **2023**, *24*, No. bbad296.
(37) Luo, S.; Chen, T.; Xu, Y.; Zheng, S.; Liu, T.-Y.; Wang, L.; He, D. One transformer can understand both 2d & 3d molecular data. In *The Eleventh International Conference on Learning Representations*, 2022.
(38) Liu, S.; Wang, H.; Liu, W.; Lasenby, J.; Guo, H.; Tang, J. Pre-training molecular graph representation with 3d geometry. *arXiv preprint arXiv:2110.07728* **2021**, No. 2110.07728.
(39) Jiang, J.; Zhang, R.; Zhao, Z.; Ma, J.; Liu, Y.; Yuan, Y.; Niu, B. MultiGran-SMILES: multi-granularity SMILES learning for molecular property prediction. *Bioinformatics* **2022**, *38*, 4573−4580.
(40) Chakrabarty, A.; Pandit, O. A.; Garain, U. Context sensitive lemmatization using two successive bidirectional gated recurrent networks. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2017; pp 1481−1491.
(41) Chen, D.; Gao, K.; Nguyen, D. D.; Chen, X.; Jiang, Y.; Wei, G.-W.; Pan, F. Algebraic graph-assisted bidirectional transformers for molecular property prediction. *Nat. Commun.* **2021**, *12*, 3521.
(42) Liang, Y.; Zhang, R.; Zhang, L.; Xie, P. Drugchat: towards enabling chatgpt-like capabilities on drug molecule graphs. *arXiv preprint arXiv:2309.03907* **2023**, No. 2309.03907.
(43) Wang, R.; Yang, M.; Shen, Y. Graph2Token: Make LLMs Understand Molecule Graphs. In *ICML 2024 Workshop on Efficient and Accessible Foundation Models for Biological Discovery*.
(44) Honda, S.; Shi, S.; Ueda, H. R. Smiles transformer: Pre-trained molecular fingerprint for low data drug discovery. *arXiv preprint arXiv:1911.04738* **2019**, No. 1911.04738.
(45) Xiong, Z.; Wang, D.; Liu, X.; Zhong, F.; Wan, X.; Li, X.; Li, Z.; Luo, X.; Chen, K.; Jiang, H.; et al. Pushing the boundaries of molecular representation for drug discovery with the graph attention mechanism. *Journal of medicinal chemistry* **2020**, *63*, 8749−8760.
(46) Liu, Y.; Wang, L.; Liu, M.; Lin, Y.; Zhang, X.; Oztekin, B.; Ji, S. Spherical message passing for 3d molecular graphs. In *International Conference on Learning Representations (ICLR)*, 2022.
(47) Hu, E. J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; Chen, W. *LoRA: Low-Rank Adaptation of Large Language Models*, 2021; https://arxiv.org/abs/2106.09685.
(48) Wu, Z.; Ramsundar, B.; Feinberg, E. N.; Gomes, J.; Geniesse, C.; Pappu, A. S.; Leswing, K.; Pande, V. MoleculeNet: a benchmark for molecular machine learning. *Chemical science* **2018**, *9*, 513−530.
(49) Guo, S.; Wang, L.; Jin, C.; Wang, J.; Peng, H.; Shi, H.; Li, W.; Guan, J.; Zhou, S. M $\Theta\{3\}$ −20M: A Large-Scale Multi-Modal Molecule Dataset for AI-driven Drug Design and Discovery. *arXiv preprint arXiv:2412.06847* **2024**, No. 2412.06847.
(50) Chen, D.; Zheng, J.; Wei, G.-W.; Pan, F. Extracting predictive representations from hundreds of millions of molecules. *journal of physical chemistry letters* **2021**, *12*, 10793−10801.
(51) Kearnes, S.; McCloskey, K.; Berndl, M.; Pande, V.; Riley, P. Molecular graph convolutions: moving beyond fingerprints. *Journal of computer-aided molecular design* **2016**, *30*, 595−608.
(52) Rong, Y.; Bian, Y.; Xu, T.; Xie, W.; Wei, Y.; Huang, W.; Huang, J. Self-supervised graph transformer on large-scale molecular data. *Adv. Neural Inf. Process. Syst.* **2020**, *33*, 12559−12571.
(53) Wang, Y.; Wang, J.; Cao, Z.; Barati Farimani, A. Molecular contrastive learning of representations via graph neural networks. *Nature Machine Intelligence* **2022**, *4*, 279−287.
(54) You, Y.; Chen, T.; Sui, Y.; Chen, T.; Wang, Z.; Shen, Y. Graph contrastive learning with augmentations. *Adv. Neural Inf. Process. Syst.* **2020**, *33*, 5812−5823.
(55) Zhang, R.; Lin, Y.; Wu, Y.; Deng, L.; Zhang, H.; Liao, M.; Peng, Y. MvMRL: a multi-view molecular representation learning method for molecular property prediction. *Briefings Bioinf.* **2024**, *25*, No. bbae298.
(56) Zhao, B.; Xu, W.; Guan, J.; Zhou, S. Molecular property prediction based on graph structure learning. *Bioinformatics* **2024**, *40*, No. btae304.
(57) Qian, C.; Tang, H.; Yang, Z.; Liang, H.; Liu, Y. Can large language models empower molecular property prediction? *arXiv preprint arXiv:2307.07443* **2023**, No. 2307.07443.
(58) Beltagy, I.; Lo, K.; Cohan, A. SciBERT: A pretrained language model for scientific text. *arXiv preprint arXiv:1903.10676* **2019**, No. 1903.10676.
(59) Yang, K.; Swanson, K.; Jin, W.; Coley, C.; Eiden, P.; Gao, H.; Guzman-Perez, A.; Hopper, T.; Kelley, B.; Mathea, M.; et al. Analyzing learned molecular representations for property prediction. *J. Chem. Inf. Model.* **2019**, *59*, 3370−3388.
(60) Ma, H.; Bian, Y.; Rong, Y.; Huang, W.; Xu, T.; Xie, W.; Ye, G.; Huang, J. Cross-dependent graph neural networks for molecular property prediction. *Bioinformatics* **2022**, *38*, 2003−2009.

https://doi.org/10.1021/acs.jcim.5c00577
J. Chem. Inf. Model. 2025, 65, 5438−5455

(61) Van der Maaten, L.; Hinton, G. Visualizing data using t-SNE. *J. Mach. Learn. Res.* **2008**, *9*, 2579−2605.
(62) Liang, J. J.; Yang, W.; Santangeli, P.; Schaller, R. D.; Supple, G. E.; Hutchinson, M. D.; Garcia, F.; Lin, D.; Dixit, S.; Epstein, A. E.; et al. Amiodarone discontinuation or dose reduction following catheter ablation for ventricular tachycardia in structural heart disease. *JACC: Clinical Electrophysiology* **2017**, *3*, 503−511.
(63) Thanvi, B.; Lo, T. Long term motor complications of levodopa: clinical features, mechanisms, and management strategies. *Postgraduate medical journal* **2004**, *80*, 452−458.

https://doi.org/10.1021/acs.jcim.5c00577
J. Chem. Inf. Model. 2025, 65, 5438−5455